

HARPER O. W. WALLACE

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EDUCATION

Universität Basel, Basel, Switzerland Nov 2022–Present
Ph.D. candidate, Biophysics

École Normale Supérieure, PSL Research University, Paris, France Sep 2020–Jun 2022
M.Sc., Cognitive Science (Neuroscience, Modelling) Rank: 3/45; GPA: 17.2/20

Denison University, Granville, OH Aug 2017–May 2020
B.S., Biochemistry, *Summa Cum Laude*, ΦBK, with Distinction Rank: 1/569; GPA: 4.00/4

RESEARCH

Theoretical and Computational Neuroscience Group Nov 2022–Present
IOB (Institute of Molecular and Clinical Ophthalmology, Basel)
Supervisor: Rava Azeredo da Silveira

Équipe biophysique et neurosciences théoriques Aug 2020–Nov 2022
LPENS (Laboratoire de Physique de l'École Normale Supérieure)
Ingénieur d'études (2021–22); *Stagiaire* (2020–21); Fulbright Scholar, France
Supervisor: Rava Azeredo da Silveira
Thesis: Efficient-coding models of context effects in perceptual decision-making

Reczek Laboratory Jan 2018–May 2020
Department of Chemistry & Biochemistry, Denison University
Supervisor: Joseph J. Reczek
Thesis: Controlled alignment of donor-acceptor columnar liquid crystals and interrogation of optical properties for applications in data storage and encryption

PUBLICATIONS

Wallace HOW, Silveira RA. Context effects arise from resource-optimal linear encoding. (In prep.)

Wallace HOW, Fikri K, Weinstein JN, Weeks WB. Improving economic conditions associated with care pattern and cost changes. *Soc Determ Health*, 7(1), **2021**. <https://doi.org/10.22037/sdh.v7i1.35144>.

Wallace HOW, Fikri K, Weinstein JN, Weeks WB. Improving economic conditions matter for mortality: Changes in local economic distress associated with mortality among Medicare fee-for-service beneficiaries between 2003 and 2015. *J Gen Intern Med*, 37(1), **2021**, 249–51. <https://doi.org/10.1007/s11606-020-06410-z>.

Van Winkle M, **Wallace HOW**, Smith N, Pomerene AT, Wood, MG, Kaehr BJ, Reczek JJ. Direct-write orientation of charge-transfer liquid crystals enables polarization-based coding and encryption. *Sci Rep–UK*, 10, **2020**. <https://doi.org/10.1038/s41598-020-72037-z>.

PATENT

U.S. Patent No. 11,151,345 (B2). Polarization-based coding/encryption using organic charge-transfer materials. Issued **2021** Oct 19. Inventors: Kaehr BJ, Reczek JJ, Van Winkle M, **Wallace HOW**. <https://patents.google.com/patent/US20210165984A1>.

PRESENTATIONS

Wallace HOW, Silveira RA. Robust decision aggregation in finite populations. Oral presentation at the Summer School on Cognitive Foundations of Decision-Making; **2025** Jul 1; Gent, Belgium.

Wallace HOW, Silveira RA. Context effects arise from resource-optimal linear encoding. Oral presentation at the Basel Neuroscience Workshop; **2024** Sep 12; Basel, Switzerland.

Wallace HOW, Fikri K, Weinstein JN, Weeks WB. Improving economic conditions matter for mortality among Medicare fee-for-service beneficiaries in the U.S.. (Virtual) Oral presentation at the iHEA World Congress on Health Economics; **2021** Jul 12–15.

Wallace HOW, Reczek JJ. Higher-order molecular control of self-assembling DACLCs. (Virtual) Poster session presented at Fundamental Research in Colloids, Surfaces & Nanomaterials Session; Division of Colloid and Surface Chemistry; American Chemical Society National Meeting; **2020** Mar 22–26.

Wallace HOW, Reczek JJ. Technology applications of self-assembling DACLCs: Information storage and encryption techniques. Poster session presented at: the 23rd Annual Posters on the Hill (CUR); **2019** Apr 30; Washington, DC.

ASSISTANT TEACHING

Neuroeconomic Foundations of Decision Making (graduate) Fall 2025
Instructors: Ernst Fehr, Rava Azeredo da Silveira

Organic Chemistry II (undergraduate) Fall 2019
Instructor: Sonya McKay

SUMMER SCHOOLS

Cognitive Foundations of Decision-Making, Gent, Belgium Jul 2025
NeuroBridges, Cluny, France Sep 2024
Cognitive Foundations of Decision-Making, Rabat, Morocco Jul 2024
CNeuro (Theoretical and Computational Neuroscience), Basel, Switzerland Aug 2022

SELECTED AWARDS & HONORS

Rosalind Swenson Fulbright Enrichment Award, *U.S.–France Fulbright Commission* 2021
Fulbright Research/Study Grant, France, *Fulbright U.S. Student Program* 2020
Valedictorian, Class of 2020, *Denison University*
William C. Ebaugh Award in Chemistry & Biochemistry, *Denison University*
Barry Goldwater Scholarship, *Barry Goldwater Scholarship Foundation* 2019
Department Fellow, Chemistry & Biochemistry, *Denison University*
Thomas A. Evans Award in Organic Chemistry, *Denison University*
23rd Annual Posters on the Hill, *Council on Undergraduate Research*